

# Wajahat Mahmood

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## EDUCATION

### CUNY Queens College

Expected Jan 2027

*Bachelor of Science in Computer Science*

*Queens, NY*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Databases, Software Engineering

## EXPERIENCE

### Software Engineer Intern, Robotics & Test Automation

May 2026 – Aug 2026

*rf IDEAS, Inc. (Roper Technologies)*

*Schaumburg, IL*

- Automated a 30–40 hour/week manual RFID setup and testing process with Python control software for a 6-axis robotic arm, enabling 3–12 hour unattended test runs.
- Automated card-to-reader configuration using barcode identification, CSV credential mapping, and HWG+ firmware loading, supporting tests of 50+ cards across 6 reader models, 4 angles, and 2 sides.
- Programmed robotic motion sequences for card pickup, suction feedback detection, barcode scanning, multi-speed reader approach, and data recording for card read height and read time.
- Hardened 100+ hours of unattended testing with reader-specific position mapping, suction-failure detection, per-read CSV checkpointing, and fault handling, maintaining under 2% observed overnight failures.

### Robotics Software Engineering Researcher

May 2025 – Jul 2025

*University of Michigan, Laboratory for Progress*

*Ann Arbor, MI*

- Co-authored a paper on RIX, a minimal POSIX-compliant robot operating system for robotics education, accepted to **IEEE/RSJ IROS 2026**; presented the research at ABRCMS 2025.
- Implemented a C++ teleoperation driver for a Trossen ReactorX-200 arm, mapping SDL2 game-controller input to joint-velocity commands and publishing joint states over RIX's pub/sub layer.
- Wrote GTest unit and integration tests for RIX's pub/sub middleware, validating message delivery and driver correctness across the teleoperation pipeline.
- Diagnosed and eliminated race conditions crashing the vendor driver by guarding shared state with mutexes and atomics.

## FELLOWSHIPS

### Google Software Engineering Program (G-SWEP)

Jun 2026 – Present

- Mentorship from a Google software engineer focused on data structures, algorithms, and technical interview preparation.

### CUNY Tech Prep, Software Engineering Fellow

Jul 2026 – Present

- Year-long full-stack fellowship covering TypeScript, React/Next.js, Node.js, PostgreSQL, TDD, and CI/CD.

## PROJECTS

### GlanceScribe | *Swift, Node.js, Gemini, ElevenLabs*

**Winner**, Hack Brooklyn 2026

- Built an AI medical scribe converting clinician video from Meta glasses into structured SOAP notes; reverse-engineered smart-glasses streaming protocol via Swift iOS app and Node.js backend.
- Engineered a multimodal pipeline pairing ElevenLabs transcription with Gemini vision, using schema-constrained JSON decoding so every note parses reliably.

### CUNY Connect | *Next.js, React, TypeScript, Supabase, Gemini*

CTP Hack 2026

- Shipped a LinkedIn-style social platform for CUNY students featuring authentication, profiles, real-time messaging, user search, posts, and commenting in a 6-hour hackathon.
- Implemented Gemini student recommendations explaining why students should connect, with database caching and fallback scoring, while building a mobile-first UI with Supabase Row Level Security and 1:1 chat with separate list/chat views on mobile.

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript, TypeScript, Swift, SQL

**Frameworks/Backend:** React, Next.js, Node.js, Flask, REST APIs, Socket.IO

**Systems:** Linux, Multithreading, TCP/UDP, WebSockets, pub/sub, ROS 2

**Databases/Testing:** PostgreSQL, Supabase, GTest

**Tools:** Git, GitHub, CI/CD, NumPy, Pandas, FFmpeg, SDL2